Patrick Godau

https://godaup.github.io/
github.com/godaup
>2500 citations, h-index: 17
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Citation Percentile: 97

Research Focus

Current:	**Surgical Foundation Models** **International Competition on Surgical Safety**
Ph.D.:	Deep Learning for Biomedical Imaging
Master's:	Algorithmic Randomness, Theoretical Computer Science

Education

Ph.D. in Computer Science (summa cum laude) — Jul 2025
Heidelberg University — *Heidelberg, Germany*
Dissertation: Lifelong Machine Learning for Biomedical Image Classification — *Advisor: Prof. Lena Maier-Hein*

M.Sc. in Mathematics — Apr 2019

B.Sc. in Computer Science — Oct 2018

B.Sc. in Mathematics — Oct 2015

Abitur (highest German school qualification) — Jun 2012

Research Experience

Postdoctoral Researcher — **since Jul 2025**
Division of Intelligent Medical Systems, German Cancer Research Center

- Lead development of Surgical Foundation Model on national HPC infrastructure at Gauss Center for Supercomputing.
- Organize the *ORena FOCUS Challenge* — Foreign Object Contextual Understanding for Safe Surgical AI — accepted at MICCAI 2026.

Visiting Researcher — **Jan 2026 - Mar 2026**

- *Surgical Safety Technologies, One World Trade Center, New York* (Jan – Feb 2026)
- *Stanford University, California* (Feb – Mar 2026), hosted by Prof. Serena Yeung-Levy

Group Lead — **since Jun 2022**
Division of Intelligent Medical Systems, German Cancer Research Center

- Lead one of three research groups within the division, with responsibility for mentoring 10–15 master's and doctoral researchers, coordinating research projects, and contributing to personnel decisions.
- Sit on the divisional board together with the Head of Division and the other Group Leads, contributing to the strategic direction of the Division of Intelligent Medical Systems.

Doctoral Researcher **May 2019 - Jul 2025**

Division of Intelligent Medical Systems, German Cancer Research Center

- Established AI evaluation standards for biomedical imaging through leadership in the international Metrics Reloaded initiative with collaboration partners from 18 countries; the resulting recommendations were adopted as a reference standard, cited by MLCommons in a 2025 U.S. FDA submission and integrated into NVIDIA's Project MONAI.
- Created and open-sourced the *Medical Meta Learner* Python package for medical imaging deep learning workflows, which has been used in 5 publications and trained more than 100,000 models.
- Pioneered methodology for knowledge transfer between datasets in data-scarce and privacy-sensitive environments plus a novel workflow improving model robustness to domain shifts in deployment.

Research Assistant **May 2016 - Nov 2017**

Division of Computer-Assisted Medical Interventions, German Cancer Research Center

- Developed meta-model for biomedical imaging challenges, standardizing data collection from >150 competitions.
- Conducted statistical analyses on ranking robustness of competition results, uncovering evaluation weaknesses.

Awards

BVM Award, *German Conference on Medical Image Computing*	2026
Winner of *RARE25: Recognition of Abnormalities in low-pREvalence cancer* Challenge	2025
Medical Image Analysis MICCAI 2023 Best Paper Award	2025
Winner of *SAGES Critical View of Safety Challenge* (category: generalization)	2024
MICCAI 2023 - Best Challenge Reviewer Award (2nd place)	2023
Winner of *Workflow Recognition in Endoscopic Pituitary Surgery* Challenge (instrument recognition)	2023
Winner of *CholecTriplet2022 Surgical Action Triplet Detection* Challenge (detection & classification)	2022
Winner of *EndoCV'22 Endoscopy Computer Vision Challenge* (task: polyp detection)	2022
Winner of *Gastrointestinal ImAge ANAlysis (GIANA)* Challenge (task: polyp classification)	2021
MICCAI 2020 - Best Challenge Reviewer Award (1st place)	2020
Winner of *HIDA Datathon on Grand Challenges on Climate Change*	2020

Grants & Funding

Gauss Center for Supercomputing Compute Grant, 11.56 mCoreh (approx. 160,000 GPU hours)	2026

Peer Reviewing

IEEE Transactions on Medical Imaging	since 2025
International Journal of Computer Assisted Radiology and Surgery	since 2025
MICCAI Challenges	since 2020

Teaching & Supervision Experience

Master's Thesis Supervision **since 2021**

- *Graph Neural Networks in Health Data Science* (2021)
- *Multi-task Learning in Surgical Data Science* (2022)
- *Autonomous Robotics for Surgery* (2025)
- *Self-Supervised Learning for Surgical Videos* (2026)

Graduate Seminar Organizer **Heidelberg University** **WS20/21 & WS21/22**

- Organized graduate seminar "Deep Learning in Medical Image Analysis", with curriculum covering state-of-the-art techniques in medical imaging (e.g., nnUnet, Federated Learning, Causality).
- Mentored students in critical analysis of current literature, presentation skills and providing feedback to peers.

Academic Tutor **Heidelberg University** **between 2013 - 2018**

- Tutored six semesters of undergraduate courses: *Linear Algebra, Theoretical Computer Science, Mathematical Logic*, thereby led weekly tutorial sessions.
- Maintained consistently high student evaluations for outstanding commitment.

Conferences and Presentations

Invited Talks

- *Large Models Meet Surgical Data Science Workshop, ISBI, London (April 2026)*
  From Foundation Models to the OR: Challenges in Auditing, Benchmarking, and Deploying Surgical AI
- *Surgical Safety Network Conference, Palm Beach, Florida (January 2026)*
  Surgical AI Challenges – Lessons Learned from SAGES-CVS & Others
- *CZI Virtual Cells Workshop, AI Modeling Benchmarking and Evaluation for Biology, Redwood City, California (September 2024)*
  Metrics Reloaded - A Large Team Effort

Oral Presentations

- *Int. Conference on Medical Image Computing and Computer-Assisted Intervention (2023)*
  Deployment of image analysis algorithms under prevalence shifts *(top 25/2253 reviewed submissions)*
- *Int. Conference on Medical Image Computing and Computer-Assisted Intervention (2021)*
  Task Fingerprinting for Meta Learning in Biomedical Image Analysis *(top 60/1630 reviewed submissions)*

Poster Presentations

- *German Conference on Medical Image Computing (2022)*
  Task Fingerprinting for Meta Learning in Biomedical Image Analysis
- *Medical Imaging meets NeurIPS workshop (2020)*
  Quantification of Task Similarity for Efficient Knowledge Transfer in Biomedical Image Analysis

Research Publications (Selection)

Journal Articles

- **Godau, P.**, Kalinowski, P., Christodoulou, E., Reinke, A., Tizabi, M., Ferrer, L., ... & Maier-Hein, L. (2025). *Navigating prevalence shifts in image analysis algorithm deployment.* Medical image analysis
- Maier-Hein, L., Reinke, A., **Godau, P.**, Tizabi, M. D., Buettner, F., Christodoulou, E., ... & Jäger, P. F. (2024). *Metrics reloaded: recommendations for image analysis validation.* Nature methods
- Reinke, A., Tizabi, M. D., Baumgartner, M., Eisenmann, M., Heckmann-Nötzel, D., Kavur, A. E., ..., **Godau, P.**, ... & Maier-Hein, L. (2024). *Understanding metric-related pitfalls in image analysis validation.* Nature methods
- Maier-Hein, L., Wagner, M., Ross, T., Reinke, A., Bodenstedt, S., Full, P. M., ..., **Scholz, P.**, ... & Müller-Stich, B. P. (2021). *Heidelberg colorectal data set for surgical data science in the sensor operating room.* Scientific data
- Maier-Hein, L., Eisenmann, M., Reinke, A., Onogur, S., Stankovic, M., **Scholz, P.**, ... & Kopp-Schneider, A. (2018). *Why rankings of biomedical image analysis competitions should be interpreted with care.* Nature communications

Conference Papers

- **Godau, P.**, Kalinowski, P., Christodoulou, E., Reinke, A., Tizabi, M., Ferrer, L., ... & Maier-Hein, L. (2023). *Deployment of image analysis algorithms under prevalence shifts.* International Conference on Medical Image Computing and Computer-Assisted Intervention
- Yamlahi, A., Tran, T. N., **Godau, P.**, Schellenberg, M., Michael, D., Smidt, F. H., ... & Maier-Hein, L. (2023). *Self-distillation for surgical action recognition.* International Conference on Medical Image Computing and Computer-Assisted Intervention
- Eisenmann, M., Reinke, A., Weru, V., Tizabi, M. D., Isensee, F., Adler, T. J., ..., **Godau, P.**, ... & Maier-Hein, L. (2023). *Why is the winner the best?* IEEE/CVF Conference on Computer Vision and Pattern Recognition
- Tran, T. N., Adler, T. J., Yamlahi, A., Christodoulou, E., **Godau, P.**, Reinke, A., ... & Maier-Hein, L. (2023). *Sources of performance variability in deep learning-based polyp detection.* International Conference on Information Processing in Computer-Assisted Interventions
- **Godau, P.**, & Maier-Hein, L. (2021). *Task Fingerprinting for Meta Learning in Biomedical Image Analysis.* International Conference on Medical Image Computing and Computer-Assisted Intervention

Languages & Skills

Machine Learning	Deep Learning, Foundation Models, Transfer Learning, Computer Vision (image classification, semantic segmentation), Lifelong & Meta-Learning, Hyperparameter Optimization, Model Validation
Programming	Proficient in Python, familiar with Java, C++, JavaScript / HTML / CSS, SQL, Haskell
Tools and Frameworks	PyTorch, Git, GitHub & CI/CD, AWS, Scikit-learn, Pandas, NumPy, Plotly, Pytest, Docker
Languages	English (proficient), German (native), French (basic)